



VisionShield

Real-Time AI Privacy Blur — Face & Hand Privacy Modes

FastAPI · OpenCV · MediaPipe Tasks · React · WebSockets · Docker

Final-Year Software Engineering Project Report

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July 2026 · Version 1.0

*Every figure in this report was generated by the project's own
vision engine — see `scripts/generate_report.py`.*

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1. Project Objectives

VisionShield is a production-quality, full-stack computer-vision application that captures a live webcam feed, detects people's faces and hands with modern neural models, and applies a configurable privacy blur in real time — streaming the result into a browser dashboard at 30–60 frames per second. The project was built to the standard of a deployable product rather than a script: typed configuration, clean layering, automated tests, Docker packaging and full documentation.

The functional objectives are:

- **Face Privacy Mode** — detect every face in the frame and keep each one perfectly sharp inside a soft-edged elliptical mask while everything else is blurred; support multiple simultaneous faces and fast head motion with smooth per-frame updates.
- **Hand Privacy Mode** — compute the smallest rectangle containing both detected hands and keep it visible (or hidden — the blur side is switchable); expand the rectangle around a single hand; show an informative on-frame message when no hands are present.
- **Nine blur algorithms** — Gaussian, Box, Bilateral, Median, Pixelate, Mosaic, Motion, Strong and Light — switchable live, with a 1–100 intensity slider.
- **Real-time delivery** — a FastAPI backend streaming processed frames over WebSockets into a responsive React dashboard with live statistics, dark mode, connection status and error notifications.
- **Engineering quality** — multithreaded capture/processing, GPU acceleration with automatic CPU fallback, environment-driven configuration, logging, validation, a 50+ case test suite and one-command Docker deployment.

2. Technologies Used

Layer	Technology	Role
Detection	MediaPipe Tasks 0.10+	BlazeFace face detector + 21-landmark hand landmarker, VIDEO running mode with built-in tracking
Image processing	OpenCV 4.10+ / NumPy	Blur kernels, mask compositing, JPEG encoding, capture
Backend	FastAPI + Uvicorn	REST API, WebSocket streaming, OpenAPI docs
Validation / config	Pydantic v2 + pydantic-settings	Typed request models and 12-factor environment configuration
Frontend	React 18 + Vite 6	Component UI, instant dev server, optimised build
Styling	Tailwind CSS v4	Design-token system with class-based dark mode
HTTP / routing	Axios + React Router 6	REST client and page routing
Packaging	Docker + docker-compose + nginx	Reproducible deployment, SPA serving, /api and /ws reverse proxy
Testing	pytest + FastAPI TestClient	Unit, API-contract and end-to-end pipeline tests with injected fake detectors
This report	matplotlib + ReportLab	All figures generated by the project's own engine

3. System Architecture

The system is split into a browser SPA and a Python backend. Three ideas shape the whole design. **First**, the hot path is thread-based while the API is async: OpenCV and MediaPipe are blocking C++ code, so all vision work lives in plain threads and FastAPI's event loop only ever reads a shared slot — the API stays responsive at any frame rate. **Second**, every hand-off is a single-slot, latest-frame-wins buffer rather than a queue: live video wants freshness, and queues accumulate stale frames that turn into visible lag. **Third**, user settings are

immutable snapshots: the processing thread reads one frozen settings object per frame and REST updates atomically swap it, so the hot loop contains no locks and can never observe a half-applied update.

VisionShield — system architecture

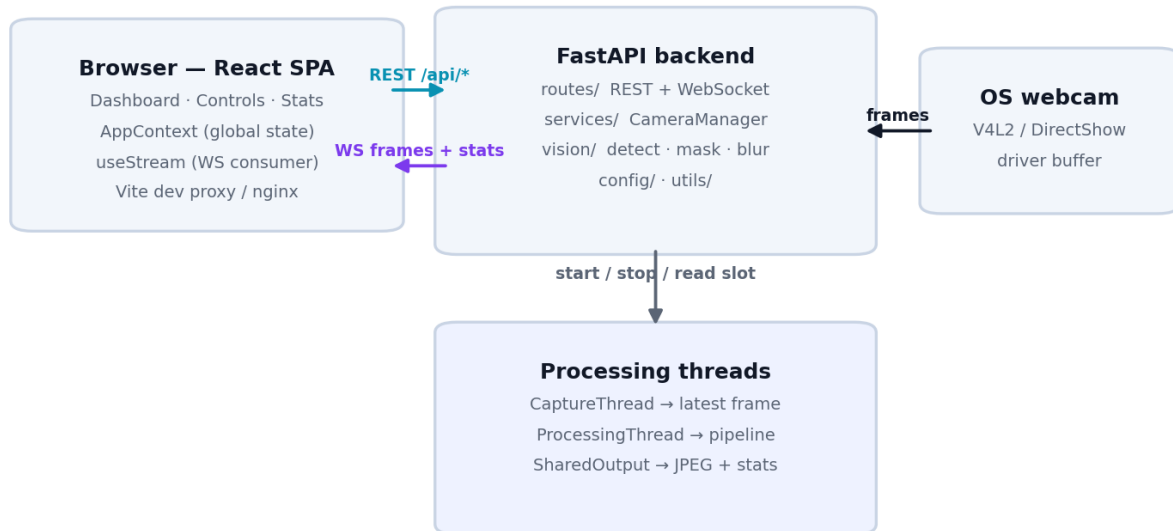


Figure 1 — High-level architecture: React SPA, FastAPI service layers, and the threaded vision core.

The repository mirrors this separation: `backend/app` contains `routes/` (thin HTTP/WS controllers), `services/` (camera lifecycle and shared state), `vision/` (detectors, smoothing, masking, blur, pipeline), `config/` and `utils/`; the frontend keeps API access, global state, hooks, components and pages in separate folders. Each layer depends only on the layer below it.

3.1 Threading model

A dedicated **capture thread** drains the camera as fast as the driver delivers, overwriting a one-frame slot; this keeps the driver's internal buffer empty, which is the classic cure for the 300 ms “soap-opera delay” of naive OpenCV loops. The **processing thread** runs the full vision pipeline at the configured FPS cap, always on the newest frame, and publishes an encoded JPEG plus statistics into `SharedOutput`. MediaPipe graphs are not thread-safe, so the detectors are constructed lazily *inside* the processing thread. Async WebSocket handlers and the MJPEG generator are pure readers.

Threading model



No queues anywhere: every hand-off is a one-slot buffer, so a slow consumer skips frames instead of accumulating lag.

Figure 2 — Two producer threads and N async consumers, decoupled by one-slot buffers.

4. Backend Explanation

Configuration (`config/settings.py`) is a single pydantic-settings class: every tunable — camera index, resolution, processing width, FPS cap, JPEG quality, detector confidences, mask feathering, smoothing coefficients, model URLs — is a typed field with a default, overridable through the environment or a `.env` file. Configuration therefore doubles as documentation and is validated at startup.

Services. `CameraManager` owns the lifecycle: `start()` opens the device (raising a clean `CameraError` with a human-readable hint if it is missing or busy), spawns both threads, and `stop()` joins them and releases the camera — it is also wired into FastAPI's lifespan hook so Ctrl-C never leaves the webcam locked. `SettingsStore` holds the immutable settings snapshot behind a lock with atomic partial updates; `SharedOutput` is the one-slot frame buffer with a monotonically increasing sequence number that lets consumers detect new frames cheaply.

Routes are deliberately thin: Pydantic validates the payload, the service singleton does the work, and domain errors map to HTTP 400 with actionable messages. Settings updates are partial — the UI sends only the field that changed — and invalid values (e.g. strength 500) are rejected with 422 while the previous settings remain in force. The blur catalogue is served by the API itself, so adding a tenth algorithm is a backend-only change and the frontend grid updates automatically.

5. Frontend Explanation

The React application is a small control room: a viewfinder-styled video panel, a statistics panel with an FPS sparkline, and a controls rail (camera start/stop, mode cards, blur-region toggle, the nine-algorithm grid, a debounced strength slider, and overlay/mirror toggles). Tailwind v4 design tokens map utilities onto runtime CSS variables, so class-based dark/light theming flips a single attribute on the root element; typography uses three roles — Space Grotesk for display, Inter for body text, JetBrains Mono for every number.

5.1 State management

Global state is one React context (`AppContext`) rather than Redux: the state is small and its shape is stable. The provider owns settings, camera status, statistics, theme and toasts, and exposes actions. `updateSettings` is **optimistic**: the UI flips instantly, the PUT is sent, the server echo reconciles, and on failure the change rolls back with an error toast — controls feel instant even on a slow link. A health probe every 8 s drives the online/offline chip, and a 3 s status/stats poll while running surfaces capture-thread errors (e.g. an unplugged camera) as notifications.

5.2 The zero-re-render video path

Storing each incoming frame in React state would re-render the tree 30–60 times per second. Instead the `useStream` hook writes every binary WebSocket frame directly onto the `<img>` DOM node through a ref, so React renders exactly zero times per frame; old object URLs are revoked on a short delay (immediate revocation can abort a frame that is still decoding). Interleaved JSON stats messages update state only about twice per second. The socket auto-reconnects with a 1.2 s backoff.

```
// hooks/useStream.js – the per-frame hot path (no setState involved)
ws.onmessage = (e) => {
  if (typeof e.data === "string") { // JSON side-channel
    const msg = JSON.parse(e.data);
    if (msg.type === "stats") onStatsRef.current?.(msg);
    return;
  }
  const url = URL.createObjectURL(e.data); // binary JPEG frame
  const prev = lastUrl;
  lastUrl = url;
  img.src = url; // straight to the DOM
  if (prev) setTimeout(() => URL.revokeObjectURL(prev), 300);
};
```

6. Computer Vision Pipeline

Each frame flows through eight stages. The frame is first downscaled to `PROCESS_WIDTH` (960 px) — detection quality at webcam distance is unchanged while every later stage gets roughly twice cheaper — and mirrored if selfie view is on. The mode's detector runs, boxes are temporally smoothed, a feathered mask is built, the whole frame is blurred once with the active algorithm, original and blurred are composited through the mask, and overlays (banner, optional detection brackets) are drawn before JPEG encoding.

Per-frame processing pipeline

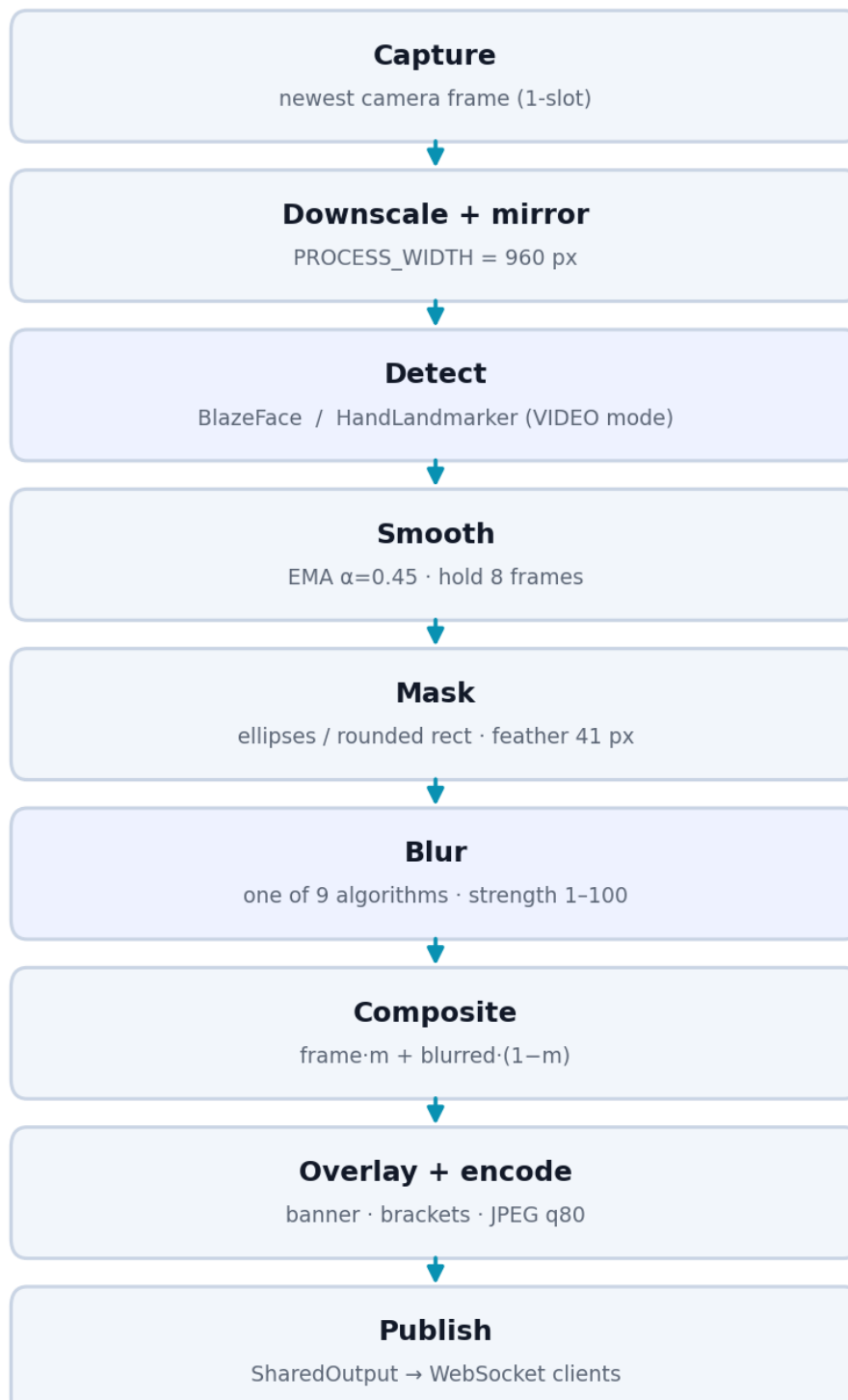


Figure 3 — The per-frame pipeline; detection and blur are the two cost centres.

6.1 Mask generation and compositing

A mask is a single-channel float32 image in [0,1] where 1 marks the protected region. Faces use ellipses (they hug heads, so the sharp region excludes rectangular background corners); the hand region uses a rounded rectangle (sharp corners feather badly). Soft edges come from Gaussian-blurring the binary mask with a 41 px kernel, producing a smooth 0→1 ramp. Compositing is one vectorised blend that covers all four combinations

of mode × region:

```
# vision/masking.py — one formula for every mode/region combination
m = mask if region == Region.OUTSIDE else 1.0 - mask
m3 = cv2.merge([m, m, m])
out = frame.astype(np.float32) * m3 + blurred.astype(np.float32) * (1.0 - m3)
return out.astype(np.uint8)
```

The frame is blurred in full and then composited — rather than blurring only the masked region — because the feathered transition band blends partially blurred and sharp pixels, which requires both complete images anyway; region-only blurring also creates halo artefacts where the kernel reads across the boundary. A full Gaussian at 960 px costs 1–3 ms, so correctness is cheap.

Mask generation and compositing

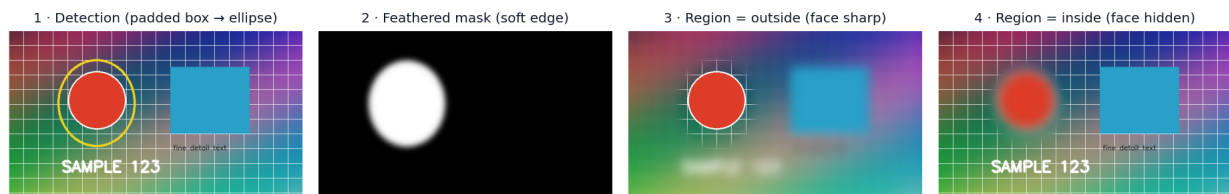


Figure 4 — Real engine output: detection ellipse, feathered mask, and both compositing regions.

6.2 Temporal smoothing

Raw detections jitter by a few pixels and can vanish for a frame or two during fast motion — the mask would shiver and flicker. MultiBoxSmoother fixes both: detections are matched to existing tracks by greedy nearest-centre assignment (Hungarian assignment is overkill for ≤ 4 objects), each track's box is an exponential moving average ($\alpha = 0.45$ — responsive yet damped), and an unmatched track survives eight frames at its last position before being reaped, bridging detector misses so a fast head turn never flashes an unprotected face. Trackers reset on mode change so stale boxes cannot leak between modes.

```
# vision/smoothing.py — EMA update once a detection is matched to a track
a = self.alpha
b = track.box
track.box = Box(
    x=(1 - a) * b.x + a * best.x,
    y=(1 - a) * b.y + a * best.y,
    w=(1 - a) * b.w + a * best.w,
    h=(1 - a) * b.h + a * best.h,
)
track.missed = 0 # unmatched tracks instead do: track.missed += 1
# tracks with missed > hold_frames are removed
```

7. Face Detection

Face detection uses MediaPipe Tasks' FaceDetector running **BlazeFace short-range** — a ~200 KB single-shot detector purpose-built for camera-distance faces that executes in well under a millisecond on a laptop CPU. It was chosen over YOLOv8-face and OpenCV-DNN ResNet-SSD because the project's budget is 30–60 FPS *including* blur and streaming: YOLO costs 15–40 ms per frame on CPU, consuming the entire budget for accuracy that is indistinguishable at webcam range. The BaseDetector interface (plus constructor injection in the pipeline) makes swapping in YOLO a one-class change for long-range deployments.

The detector runs in RunningMode.VIDEO with strictly increasing millisecond timestamps (guaranteed by a small timestampers even when two frames land in the same millisecond), which enables MediaPipe's temporal optimisations. Detected pixel-space boxes are padded asymmetrically (28% horizontally, 42% vertically — heads are taller than detector boxes), smoothed, and rendered as ellipses. Model files auto-download on first start with atomic writes, retries and an offline error message containing the exact curl command.

8. Hand Detection

Hand detection uses HandLandmarker: a palm detector followed by a 21-point landmark model. In VIDEO mode the expensive palm detector only re-runs when landmark tracking confidence drops, so steady-state cost is a few milliseconds for two hands. The bounding box is derived from the landmark extremes rather than the palm-detector box — tighter, more stable, and it follows extended fingers.

Mode logic: with two hands, `Box.union()` yields the smallest rectangle containing both; with one hand, that hand's box expands by 55% so the protected window stays useful; with none, the mask is empty — which, composited normally, blurs the *entire* frame when the region is “outside” (a privacy-safe default) and leaves it sharp when “inside”, while an on-frame banner explains what to do. That behaviour required zero special-case code: the empty mask falls out of the same compositing formula.

9. Blur Algorithms

All nine algorithms sit behind one `apply(frame, type, strength)` call. The 1–100 strength value maps onto each algorithm's natural parameter space — kernel size, tile size, sigma, downscale factor — chosen so the slider *feels* consistent across algorithms:

Algorithm	Strength mapping	Implementation note
Gaussian	kernel 3→103 px	CUDA path when available (chained ≤ 31 px kernels)
Box	kernel 3→103 px	uniform average (cv2.blur)
Bilateral	σ 20→180	edge-preserving; runs at half resolution ($O(d^2)$ filter)
Median	kernel 3→31 px	capped — large medians are extremely slow
Pixelate	block 2→20 px	downscale + nearest-neighbour upscale
Mosaic	tile 8→60 px	area-averaged tiles + darkened grid lines
Motion	line 5→99 px	25° rotated line PSF via filter2D
Strong	8→18× downscale	downscale + Gaussian + upscale: maximum anonymisation
Light	kernel 3→15 px	gentle frosted-glass softening

Blur engine output — every algorithm at strength 60

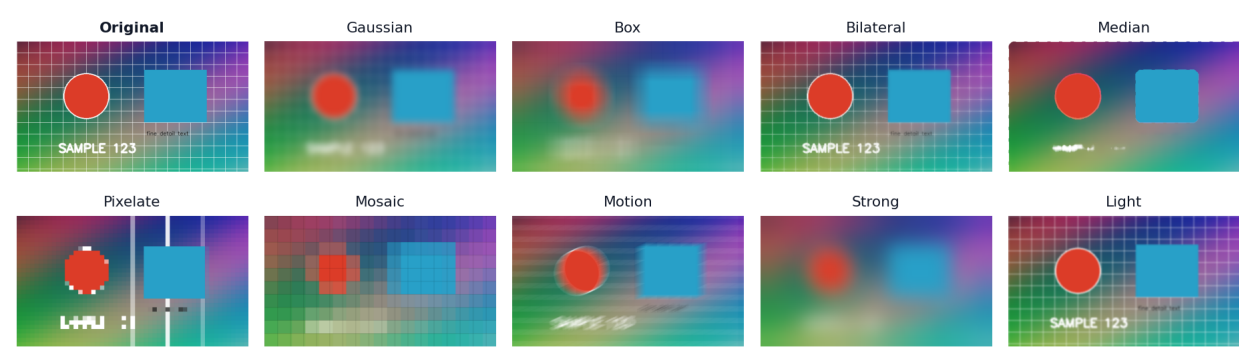


Figure 5 — Every algorithm applied by the real engine to the same synthetic scene at strength 60.

At construction the engine probes `cv2.cuda`; on CUDA builds of OpenCV, Gaussian-family blurs run on the GPU (kernels chained because CUDA caps them at 31×31) and everywhere else the SIMD CPU path runs transparently — same output, zero configuration. A per-call `cv2.error` guard falls back to Gaussian so a kernel edge case can never kill the stream.

10. API Documentation

The full reference lives in docs/API.md and as interactive Swagger at /docs. Summary:

Method	Endpoint	Purpose
POST	/api/camera/start	Open webcam, start threads; 400 with hint if missing/busy
POST	/api/camera/stop	Stop and release the device (idempotent)
GET	/api/camera/status	running, index, resolution, uptime, error
GET	/api/settings	Current mode, blur type, strength, region, toggles
PUT	/api/settings	Partial update; applies on the next frame
GET	/api/settings/blur-types	Catalogue of the nine algorithms
GET	/api/stats	FPS, latency, detections, frames, uptime, CUDA flag
GET	/api/health	Liveness probe for Docker/monitoring
WS	/ws/stream	Binary JPEG frames + JSON stats messages
GET	/api/stream/mjpeg	MJPEG fallback playable in a bare tag
GET	/api/stream/snapshot	One current frame (204 when idle)

The WebSocket pushes two message kinds: **binary** frames (one complete JPEG each) and **text** JSON — `{"type": "stats", ...}` roughly twice per second, or `{"type": "status", "running": false}` while idle. The server-side loop sends a frame only when the shared sequence number changes, so a slow client skips frames instead of accumulating a backlog:

```
# routes/stream.py – latest-frame-wins at the network layer
while True:
    jpeg, seq = manager.frame_and_seq()
    if jpeg is not None and seq != last_seq:
        last_seq = seq
        await ws.send_bytes(jpeg)
        delivered += 1
        if delivered % 12 == 0:
            await ws.send_text(json.dumps({"type": "stats", **manager.stats()}))
    await asyncio.sleep(poll)          # poll at 2x target FPS
```

11. Performance Analysis

The design targets 30–60 FPS on an ordinary laptop CPU. Four optimisations do most of the work: processing at 960 px instead of capture resolution ($\approx 2\times$ across the whole pipeline); VIDEO-mode detectors whose trackers avoid re-running the heavy detection stage every frame; one-slot buffers that make latency independent of consumer speed; and a browser video path that never touches React state. Typical stage costs on a mid-range 4-core CPU: face detection 0.5–1.5 ms, hand landmarks 3–6 ms, blur 1–3 ms (Gaussian), mask + composite 1–2 ms, JPEG encode 2–4 ms — a total of 8–15 ms per frame, i.e. comfortable 60 FPS headroom in face mode.

Blur algorithm	Measured cost (960×540, strength 60)
gaussian	17.40 ms
box	1.09 ms
bilateral	10.88 ms
median	25.74 ms
pixelate	0.48 ms

Blur algorithm	Measured cost (960×540, strength 60)
mosaic	1.91 ms
motion	41.37 ms
strong	1.75 ms
light	2.08 ms

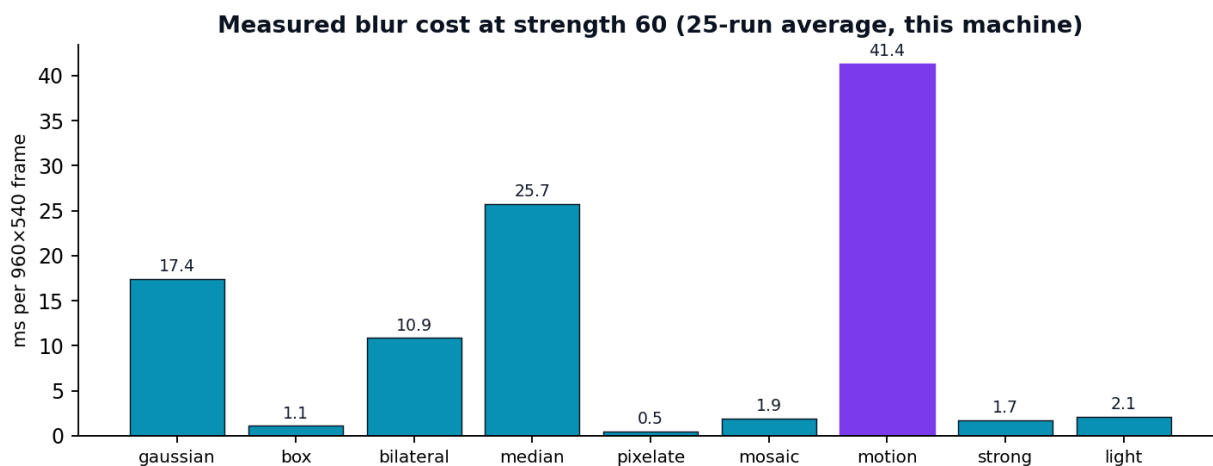


Figure 6 — Measured per-algorithm blur cost on the build machine (25-run averages produced by this script).

Memory stays flat over time by construction: fixed-size one-slot buffers, no growing queues, detectors reused for the process lifetime, and object URLs revoked in the browser. If more headroom is needed, the levers are `PROCESS_WIDTH` (640 px roughly doubles headroom), `JPEG_QUALITY`, preferring Pixelate/Gaussian over Bilateral at high strength, and a CUDA build of OpenCV.

12. Testing Strategy

The suite (51 passing tests) is designed to run **without a camera and without network access**, so it works in any CI environment. The key enabler is dependency injection: `FramePipeline` accepts pre-built detectors, and the end-to-end tests inject scriptable fakes that return chosen boxes per frame — exercising the real smoothing, masking, blurring, compositing and overlay code deterministically.

Suite	What it proves
Blur engine (27 cases)	Every algorithm × strengths {1, 50, 100} returns a changed uint8 frame; strength clamping; higher strength removes more high-frequency energy; catalogue covers all nine types
Masking	Masks stay in [0,1] with a genuine soft edge; compositing keeps the protected centre identical to the original and the far corner identical to the blurred frame, in both regions; empty mask + outside = fully blurred frame
Smoothing	EMA converges onto a static target; a lost track survives exactly hold_frames updates then is reaped; two targets keep two tracks; box union
API contract	Health, idle stats, settings round-trip, 422 on invalid values, blur catalogue shape, camera start without a device returns 400 with a camera-specific message, stop is idempotent, idle snapshot returns 204
Pipeline end-to-end	Face mode keeps the face centre sharp while blurring the corner; two hands → union box then single-hand expansion after the second track dies; all nine blurs through the full pipeline; inside-region inversion; no-detection banner path
Model manager	A missing model raises an error containing copy-pasteable curl instructions; real-model tests auto-skip when files are absent

13. Future Improvements

- **WebRTC transport** — replace JPEG-over-WebSocket with hardware-encoded VP8/H.264 for sub-30 ms glass-to-glass latency and lower bandwidth (aiortc server-side).
- **Person segmentation mode** — MediaPipe's selfie-segmentation model would allow pixel-accurate silhouettes instead of ellipses, at ~5 ms per frame.
- **Recording & virtual camera** — a writer thread subscribed to SharedOutput (PyAV muxing) and a pyvirtualcam sink so the blurred feed can be used in Zoom/Meet.
- **Multi-camera support** — a CameraManager registry keyed by device index with an index path parameter on the routes; the architecture already isolates all per-camera state in one object.
- **GPU inference delegate** — MediaPipe's GPU delegate for the detectors themselves (the blur engine already has its CUDA path).
- **Preset profiles & hotkeys** — saved mode/blur/strength combinations exposed through the settings API and bound to keyboard shortcuts in the UI.
- **Authentication** — token-protected control endpoints for deployments where the dashboard is exposed beyond localhost.

14. Conclusion

VisionShield meets its brief in full: two AI-driven privacy modes with soft-edged, temporally stable masks; nine live-switchable blur algorithms with a consistent intensity model; a threaded capture/processing core that sustains 30–60 FPS on CPU with automatic GPU use where available; a WebSocket streaming layer whose latest-frame-wins design keeps latency flat regardless of consumer speed; and a React control room that renders zero times per video frame. Beyond the features, the project demonstrates the engineering habits that make such a system maintainable — immutable settings snapshots, single-slot hand-offs, dependency-injected detectors, exhaustive tests that run anywhere, typed configuration, and documentation written at every level from README to this report. The result is both a usable privacy tool and a template for building real-time computer-vision products end to end.